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PUBLIC SERVICE MOTIVATION, RELATIONAL JOB DESIGN, AND JOB SATISFACTION IN LOCAL GOVERNMENT

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This study seeks to examine whether and how a group of Australian local government workers' perceptions of two relational job characteristics – impact on citizens and contact with citizens – influence the relationship between public service motivation (PSM) and job satisfaction. Government employees who have strong norms about performing public service were found to be more satisfied with their job partly based on the extent to which they perceived that their jobs provided avenues for worthwhile accomplishment through the frequency, magnitude, and scope of job impact. In contrast, the employees' perceptions of the frequency and breadth of contact with the recipients of their service did not significantly influence PSM, or the association between PSM and job satisfaction.

INTRODUCTION

This research aims to examine whether and how perceptions of two relational job characteristics – the impact of their job on service recipients (henceforth referred to as job impact) and the interactions with service recipients (job contact) – positively influence the relationship between public service motivation (PSM) and job satisfaction in a group of Australian local government workers. Research on the organizational consequences of PSM has recently progressed to identifying organizational factors that moderate and to a lesser extent mediate the relationships between PSM and outcomes (Bright 2007; Taylor 2008; Grant and Sumanth 2009; Vandenabeele 2009). In the process, these studies have presented different views on the efficacy of these organizational factors and how they work.

In particular, the US studies conducted by Grant and colleagues have shown that jobs could be relationally structured to foster prosocial motivation, a concept that is related to PSM (Morgeson and Humphrey 2006; Grant 2007; Grant and Parker 2009). This implies that jobs that are capable of making employees' public service values more salient and provide opportunities for them to fulfil these values by interacting with service recipients and witnessing the positive impact they have made on the service recipients may be all that differentially motivate and satisfy high PSM employees from low PSM employees. Yet other literatures, such as Lipsky's (1980) work on street-level bureaucrats, have described the experience of interacting with service recipients to be fraught with difficulties and tensions.

Different studies have also shown different ways in which relational job characteristics can affect the relationship between PSM and a job outcome. Taking job impact as an example, Grant and Sumanth (2009) reported that it moderated the association between PSM and job performance, but Vinzant (1998) noted that it mediated the relationship between PSM and job satisfaction. These different views of the extent to and process by which organizational factors shape PSM's effects on job outcomes highlight the need for not just more research to be conducted in this area, but more importantly, for a framework

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to be developed that explains why and how these organizational factors influence PSM's impact on job outcomes.

This study can thus contribute to the public administration scholarship in two ways. The first is its theoretical contribution. It draws from multiple theories – March and Olsen's (1989) 'logic of appropriateness', several self-regulation theories, and Perry's (2000) process theory – to develop a framework that proposes why and how organizational factors interact with employees' PSM to affect a job outcome such as job satisfaction (Locke 1976; Wright and Kim 2004). Rather than assume a specific pathway, this framework allows for mediating and moderating effects to be tested in order to determine whether there is more than one mechanism by which the two relational job characteristics intervene in the relationship between PSM and job satisfaction.

The second contribution of this study is its findings. Although there is progressively more PSM research conducted outside the USA, particularly in Europe, a non-US study such as this that examines the intervening effects of job impact and job contact on the relationship between PSM and job satisfaction has not been attempted before. Given the difference in context, it is unknown whether these intervening organizational factors would operate in the same way, or even whether they would work in Australia. Issues such as these must be addressed if PSM research is to reach its full potential. In addition, studies on relational job characteristics are often more general or focused than comprehensive (Grant 2008a; Grant and Sumanth 2009; exception: Grant 2008b).

Yet, perceived job impact can vary in terms of frequency (the number of times the job provides opportunities to positively affect citizens), magnitude (the extent and duration of potential positive effects on citizens), and scope (the number or range of people potentially affected). Similarly, perceived job contact can vary along frequency (the number of times the job provides opportunities to interact with citizens), and breadth (the extent to which the job provides opportunities to interact with different citizens) (Grant 2008b). This study distinguishes itself from others in recognizing these finer qualities of relational job characteristics. Finally, if PSM can be influenced by organizational factors, then the mediation and moderation research findings can help to inform strategies that managers can use to foster PSM and enhance its impact on job satisfaction.

This article is organized as follows. In the next section, the relevant theories supporting the influence of organizational factors in shaping the relationships between PSM and job outcomes are discussed. This is followed by a literature review of perceived relational job characteristics as potential mediators and moderators in the PSM–job satisfaction relationship. Subsequent sections present the methodology, results, and discussion. The latter section also discusses the limitations of the study and presents recommendations for future research.

INTERACTIONS BETWEEN PSM AND ORGANIZATIONAL FACTORS: THEORETICAL BASES

The interactions between employees' PSM and organizational factors in determining job outcomes are supported by various theories. They include March and Olsen's (1989) 'logic of appropriateness', self-regulation theories, and Perry's (2000) process theory.

March and Olsen's (1989) model of motivation associated with the logic of appropriateness recognizes that employees' identity factors (e.g., prosocial motives or PSM, and personality dispositions and experiences) can interact with situational cues, and play a major role in determining the employees' definition of the situation they face in their

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organization. Employees' decisions are said to result from them asking, 'What does a person like me do in a situation like this?' (Kelley and Thibaut 1978; Weber *et al.* 2004). The appropriateness framework highlights the importance of not only PSM in shaping job attitudinal outcomes, but also PSM's interaction with situational cues in influencing these outcomes. If employees view situations in their organization through a lens constituted by the interaction between their public service motives and situational cues, then it suggests that situational cues in their organization can influence the relationship between PSM and job attitudes such as job satisfaction.

Similarly, self-regulation theories, such as the social cognitive theory and self-determination theory, propose that employees' self-regulation can influence their cognitions and in turn behaviour (Bandura 1986; Ryan and Deci 2003). According to Bandura (1986), self-regulation is a function of three sub-functions, self-observation, judgmental processes, and self-reaction. Employees engage in self-monitoring in order to influence their own behaviour. The capacity of self-monitoring to influence motivation is dependent upon the employees' capabilities to judge their self-observed behaviour against a set of internal standards. These standards could be derived from social and cultural cues, and the evaluative standards developed by others. Weber *et al.* (2004) also applied the logic of appropriateness to explain that employees' prosocial outcomes could be partly due to their tendencies to self-monitor. Strong cooperators reported high scores on self-monitoring activities, and high self-monitors reported cooperating more in social situations than low self-monitors (De Cremer *et al.* 2001; Kurzban and Houser 2001).

Finally, Perry (2000) proposed a process theory of PSM. He identified three sets of factors that could interact to affect employees' behaviour: (1) the employees' environment outside their organization, namely the sociohistorical context, which covers education, socialization, and life events; (2) the employees' motivational context or their organization, for example the work environment, organizational beliefs, values and ideologies, organizational incentives, and job characteristics; and (3) individual characteristics, such as the employees' abilities, self-concept, and self-regulatory processes. In conclusion, theories such as the above suggest that rather than a sole direct impact on job satisfaction, employees' PSM can interact with organizational factors to affect job satisfaction.

If organizational factors interact with PSM to affect job satisfaction, then it is next relevant to understand the mechanism. Do they function largely as a mediator and/or moderator? Some self-regulation theories such as the predisposition–opportunity theory and person–organization fit theory support a mediating effect (Knoke and Wright-Isak 1982; Kristof 1996). In the predisposition–opportunity theory, Knoke and Wright-Isak (1982) argued that the self-regulation of employees' job outcome depends on the incentives offered by the organization. They proposed that a match between the incentive systems and individual motives contributes to the employee's commitment to the organization and in turn affects his/her job outcome. Without a match between organizational incentives and individual motives, no self-regulated public service motivated behaviour will occur (Perry and Vandenabeele 2008). Similarly, the person–organization (P–O) fit literature maintains that self-regulation only occurs if there is a match between the characteristics of the individual and those of the organization (Kristof 1996; Verquer *et al.* 2003). Without a match, no motivation and therefore no desirable outcome will eventuate. It suggests that employees whose values and goals are compatible with those of their organization will have more favourable job satisfaction than their counterparts who are less compatible.

According to Wright and Pandey (2008), the mediating effect of P–O fit on the PSM–job satisfaction relationship occurs in two stages. First, employees with high PSM levels are

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more likely to agree with the service mission and objectives of public organization that are charged with performing public service. Second, the more employees agree with their organization's mission and objectives, the more likely they would find the tasks assigned by the organization satisfying, particularly when they provide an opportunity for employees to have a significant positive impact on the well-being of the recipients of their work (Hackman and Oldham 1976; Grant 2007). For employees with high levels of PSM, the P–O fit literature suggests that high job satisfaction is dependent on the employees' conviction that their organizational tasks allow the realization of their public service motives. Failure to allow this will result in lower job satisfaction.

Other self-regulation theories, notably self-determination theory, suggest a moderating effect of organizational factors. Ryan and Deci's (2003) framework shows a continuum of five different motivations, ranging from controlled to autonomous motivation. On the one hand, identities are acquired when one is naturally interested in the activities these identities bring about. Behaviours that are linked with these identities are considered to be intrinsically motivated (intrinsic regulation). On the other hand, one also acquires identities for reasons other than this natural inclination. Ryan and Deci (2003) stated that individuals have three basic psychological needs: the need for autonomy, the need for relatedness, and the need for competence. The internalization of values into an identity, which is a source of motivation, is based upon the satisfaction of these three needs (extrinsic regulation). Extrinsic regulation is subdivided into four categories along the continuum. The four types of identity regulation that correspond to four types of extrinsic motivation suggest that if an organization fosters the basic needs of individuals, the degree of internalization of identities will be higher. The effect of this regulation on behaviour is also stronger to the extent that the associated identities are autonomous. Based on self-determination theory, the more autonomous the employees with high PSM levels perceive their identity, the higher their job satisfaction.

Having provided the theoretical support for the mediating and moderating effects of organizational factors on the relationship between PSM and job satisfaction, the next step is to identify the organizational factors that may function as mediators and moderators. In his process theory of PSM, Perry (2000) has indicated that job characteristics and the work environment are influential organizational factors. Instead of viewing jobs solely as collections of tasks, many researchers are now recognizing that the social context of work plays a vital role in shaping employees' experiences and actions (Morgeson and Humphrey 2006; Grant and Parker 2009; Kanfer 2009). Employees' perceptions of relational job characteristics may thus mediate and/or moderate the relationship between PSM and job satisfaction.

PERCEIVED RELATIONAL JOB CHARACTERISTICS

Direct effects on PSM and job satisfaction

In his 2007 conceptual framework on the relational architecture of jobs, Grant proposed that two relational job characteristics can motivate employees to do good for service recipients. The first is a job that provides opportunities for employees to have a positive impact on service recipients (Hackman and Oldham 1976). The second is a job that provides opportunities for employees to have contact with service recipients (Gutek *et al.* 1999; Kanfer 2009). Perceptions of both aspects strengthen PSM by connecting employees to the impact of their actions on other people. Job impact signals to workers that their

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actions matter to the well-being of other people, and contact with people provides workers with access to feedback about their impact (Grant 2007).

These two relational job characteristics also affect job satisfaction. Humphrey *et al.* (2007) reported a significant relationship between different relational job characteristics (e.g., interaction outside the organization and feedback from others) and job satisfaction. When employees perceive a high degree of task significance, their job satisfaction will rise because they gain opportunities to fulfil their higher order needs, such as self-actualization and self-esteem (Wright and Kim 2004). Assuming that individuals are proactive, growth oriented, and responsive to their environment (Ryan and Deci 2003), task significance or job impact can positively affect job outcomes by allowing workers to experience their work as more meaningful (Hackman and Oldham 1976).

While job impact is proposed to be positively related to job outcomes (Hackman and Oldham 1976; Morgeson and Humphrey 2006), it is unclear whether job contact would have a similar positive impact. The relational job design literature views contact with citizens as a positive experience for employees. Here, contact with people whom they serve provides employees with access to feedback on the effects of their actions on service recipients, and feedback about how to best do their job (Humphrey *et al.* 2007; Grandey and Diamond 2010). When employees interact with the service recipients, they are also better able to understand the recipients' perspectives, and thereby identify and empathize with these recipients, which can lead them to develop stronger affective commitment to the recipients (Parker and Axtell 2001). For example, interactions with the public have been reported to be positively related to job satisfaction (Morgeson and Humphrey 2006).

In contrast, the literature on emotional labour views interactions with citizens less positively. Interactions with service recipients are argued to place high emotional demands on the employees. Employees are required to induce or suppress certain feelings to produce a state of mind or emotional reaction among the service recipients (Zapf *et al.* 2001; Grandey and Diamond 2010). Unlike the former perspective which views service recipients as a source of information, service recipients are here viewed as a source of conflict; contact with the public can involve unpleasant communication, internal conflict, and other personal costs (Grandey and Diamond 2010). In contrast to the job design perspective which emphasizes the benefits of brief and infrequent interactions, it focuses on the costs of frequent, direct, lengthy, and emotionally intense interactions with the service recipients (Cordes and Dougherty 1993).

Similarly, Lipsky's (1980) work on street-level bureaucrats sees the interactions with service recipients as being fraught with tension. Public employees who interact directly with citizens or street-level bureaucrats generally face constraints in fulfilling their expectations in their daily work that are attributed to various factors, ranging from insufficient resources to the institutional characteristics of the job (e.g., red tape). Lipsky argued that street-level bureaucrats who are unable to meet citizens' demands (quantity and substance) as a result of these barriers are likely to use coping mechanisms as a self-defence, such as providing differential treatment to different citizens.

Whether this experience would in turn strengthen or dampen job satisfaction is unclear. On the one hand, the fact that street-level bureaucrats are handling citizens differently from what they ideally desire may lower their job satisfaction levels. On the other hand, authors such as Bussing (1992, 1998) have indicated that job satisfaction can exist in different forms in accordance with the employees' response to a mismatch between their personal aspirations and their perceptions of the work environment. Resigned satisfaction is when employees believe that their work environment fails to meet their

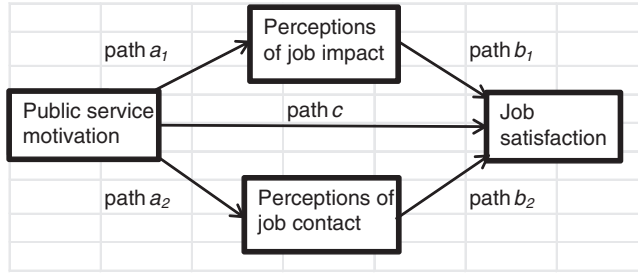
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personal expectations and they cope with this negative condition by reducing their level of aspiration in order to maintain a positive state of satisfaction (Bussing 1998; Giauque *et al.* 2011). This implies that coping mechanisms can be utilized to avoid frustration as well as to gain satisfaction (Nielsen 2006). There is thus inconclusive evidence on whether job contact with citizens, particularly the frequency and breadth of the contact experience, would strengthen, dampen, or have a negligible effect on the relationship between PSM and job satisfaction.

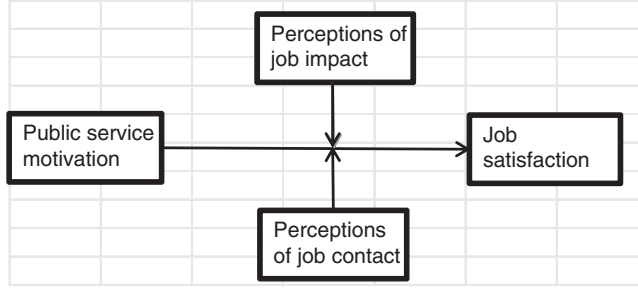
Mediation and moderation effects on the PSM–job satisfaction relationship

Due to lack of empirical evidence, this study does not posit any specific hypothesis about whether employees' perceptions of job impact (in terms of frequency, magnitude, and scope) and job contact (frequency and breadth) operate as a mediator, a moderator, neither, or both. The potential mediating and moderating effects of the job impact and job contact variables are diagrammatically summarized in figure 1.

On the one hand, perceptions of these two relational job characteristics may mediate the relationship between PSM and job satisfaction. Employees with high PSM levels are sensitive to information about the consequences of their actions on others or job impact cues (Korsgaard *et al.* 1997; Grant 2008b). They are likely to self-monitor their work progress (Weber *et al.* 2004). Based on the predisposition–opportunity theory and P–O fit theory (Knoke and Wright-Isak 1982; Kristof 1996), it is possible for employees with high PSM levels to only show high job satisfaction levels when they are convinced that their jobs allow them to have a positive impact on and/or contact with citizens.



Mediating effects



Moderating effects

FIGURE 1 *Testing the mediating and moderating effects of relational job characteristics on the PSM–job satisfaction relationship*

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On the other hand, perceptions of job impact and job contact may affect the strength of the relationship between PSM and job satisfaction. Grant and Sumanth (2009) found that perceived task significance has a positive moderating effect on the relationship between prosocial motivation and the job performance of fundraising callers in terms of the number of calls made. They argued that unlike employees with lower levels of PSM who were less concerned about doing work that benefited others, high PSM employees would raise their performance when they perceived their task to be significant because of a conviction that their job was more meaningful and helpful to and valued by the service recipients. Based on self-determination theory (Ryan and Deci 2003), it is possible for the job satisfaction of employees with high PSM levels to be higher when the tasks in their organization satisfy their basic psychological needs. Therefore, this research tests the claim that job impact and job contact operate as both a mediator and a moderator on the relationship between PSM and job satisfaction.

METHOD

Sample and procedures

The survey was targeted at Australian local government workers in permanent employment. A letter was sent to the CEOs of 11 local councils in an Australian state, seeking permission to run a survey on their employees. Seven agreed to participate.

Data were collected by an online survey, and to a lesser extent a paper-based survey. They were distributed to local government workers through a senior HR council staff member. To satisfy the university research ethical guidelines, the researcher did not directly approach the respondents, but left instructions with the HR director or senior manager of the council for the respondents to be in permanent employment and randomly selected, preferably with access to a computer. A paper-based survey was used to reach some respondents who did not have immediate access to a computer in two of the seven local councils.

A total of 247 workers submitted their responses, but 14 only completed the demographic section and did not complete the rest of the survey. Their responses were discarded, making a total of 233 useable responses. Based on an estimated total number of people approached, the response rate was determined to be about 30 per cent. Missing data were less than 5 per cent and did not show any systematic pattern. This implies that they are unlikely to pose a problem in the interpretation of results. The expectation-maximization (EM) algorithm method was used to deal with the small proportion of missing data.

A majority of the survey respondents were female (69 per cent), held a university undergraduate or postgraduate degree (51 per cent) and a supervisory position (51 per cent). On average, they were aged 41 years, and had worked in local government for nine years. Unfortunately, there was insufficient information provided by the councils to check on the representativeness of the sample group; in most instances, only statistics on gender, age, and length of service were available at an aggregate level. Permanent employees made up about 67 per cent of the workforce in the local government councils surveyed, but demographic statistics of permanent employees were unavailable, which is a limitation of this study. The general workforce of the councils surveyed was mainly female (56 per cent), and had a mean age of 39 years and a mean length of service of five years. The sample of mainly permanent employees appeared to be made up of more

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females, and represented an older group with longer length of service than the general workforce in permanent, fixed term contract, and casual employment.

Measures

The independent variable, PSM, was measured with a few items from a scale developed by Kim *et al.* (2010), which was based on the model designed by Kim and Vandenabeele (2010), and closely related to Perry and Wise's (1990) model. Noting a few limitations using Perry's (1996, 1997) scale to measure the PSM construct in non-US environments (Kim and Vandenabeele 2010), Kim *et al.*'s (2010) scale was designed to facilitate international research on PSM. This study utilized seven items from their scale to produce a unidimensional measure of PSM: (1) one item on attraction to public service, which measures the instrumental motives of PSM, and captures an individual's disposition to serve the public interest through public participation; (2) two items on commitment to public values, which assesses the value-based motives of PSM, and is related to personal beliefs or creeds; (3) two items on compassion, which determines the identification motives of PSM, and covers affective bonding with identified objects, such as other members of a social category or of a political system; and (4) two items on self-sacrifice. The study scales and items are presented in Appendix 1. Confirmatory factor analysis (CFA) using the Mplus version 6 program showed a good fit for a unidimensional scale; $\chi^2(14) = 17.33$, $p > 0.05$; Root Mean Square Error of Approximation (RMSEA) = 0.03 (range 0.00–0.08); Standardized Root Mean Square Residual (SRMR) = 0.03; Comparative Fit Index (CFI) = 0.99; and Tucker Lewis Fit (TLI) = 0.99. The overall PSM scale had a Cronbach's alpha (α) of 0.84.

The intervening variables, perceived job impact and perceived job contact, were measured with Grant's (2008b) scales on these constructs. Five items measure the frequency (two items), magnitude (one item), and scope (two items) of job impact. A one-factor congeneric model of the items showed a good fit; $\chi^2(2) = 4.14$, $p > 0.05$; RMSEA = 0.00 (0.00–0.08); SRMR = 0.02; CFI = 1.00; and TLI = 1.01 (reliability, $\alpha = 0.91$). Four items measure the frequency (two items) and breadth (two items) of job contact. Again, a CFA on a one-factor congeneric model of these items showed a good fit; $\chi^2(2) = 1.22$, $p > 0.05$; RMSEA = 0.00 (0.00–0.11); SRMR = 0.02; CFI = 1.00; and TLI = 1.02 ($\alpha = 0.95$).

The dependent variable, job satisfaction, was measured by four items from Brayfield and Rothe's (1951) Job Satisfaction Index. A CFA showed a good fit for a unidimensional model; $\chi^2(2) = 3.77$, $p > 0.05$; RMSEA = 0.06 (0.00–0.16); SRMR = 0.03; CFI = 0.97; and TLI = 0.90 ($\alpha = 0.84$).

To confirm the separateness of the scales, a factor analysis was carried out on all items. The results are found in Appendix 2.

Most empirical studies which use a self-administered survey face a risk of common method bias (CMB), and this study is not an exception. The fact that all data were collected from the same source could contribute to a response bias, such as halo effects, and social desirability, which in turn could threaten the validity of the conclusions about the relationships between measures (Podsakoff *et al.* 2003). A Harman's single-factor test was conducted to determine the extent of CMB in the current data. A factor analysis of all items from the theoretical constructs of interest produced a variance of 35 per cent, which is below the acceptable maximum threshold of 50 per cent of total variance (Podsakoff and Organ 1986). This was followed by the unmeasured latent methods factor technique (Podsakoff *et al.* 2003; Richardson *et al.* 2009). Here, the items were loaded on their theoretical constructs and a latent common methods variance factor. Based on this

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TABLE 1 *Means, standard deviations, and correlations*

Variable	Mean (S.D.)	1	2	3	4	5	6	7	8
1. PSM	5.63 (0.65)								
2. Perceived job impact	5.11 (1.16)	0.23***							
3. Perceived job contact	5.19 (1.32)	0.09	0.67***						
4. Job satisfaction	5.40 (1.02)	0.26***	0.39***	0.33***					
5. Gender	1.69 (0.46)	0.03	0.06	0.12	0.18**				
6. Age	40.5 (11.21)	0.05	0.14*	0.04	0.13	−0.19**			
7. Education	4.34 (1.32)	0.13	0.05	−0.09	−0.07	−0.20**	−0.18**		
8. Supervisory position	1.51 (0.50)	0.04	0.13*	0.08	−0.02	−0.30***	0.26***	0.21***	
9. Length of service	8.86 (8.07)	0.01	0.08	−0.02	−0.04	−0.29***	0.49***	−0.10	0.32***

Level of significance: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

technique, the common variance was determined to be 12 per cent, which is less than half the amount of method variance (a median of 25 per cent) observed by Williams *et al.* (1989).

RESULTS

Table 1 provides the descriptive statistics of the variables, and the correlations between them. On average, the respondents 'somewhat agreed' that their job has provided them with an opportunity to have a positive impact on and contact with citizens. Their PSM and job satisfaction levels fell, on average, between 'somewhat agree' and 'agree'.

Table 1 also shows a significant association between the main study variables. PSM was positively related to job satisfaction ($r = 0.26$, $p < 00.001$). PSM was also positively associated with job impact on citizens ($r = 0.24$, $p < 00.01$), but not job contact with citizens. Job satisfaction was positively related to job impact ($r = .39$, $p < 0.001$) and job contact ($r = .33$, $p < 0.001$).

To test for the mediating and moderating effects, the procedures outlined by Baron and Kenny (1986) are followed. Mediation is determined in four steps. First, the independent variable (PSM) should be significantly correlated with the dependent variable and (job satisfaction). The results of this total effect of the PSM variable on the job satisfaction variable are displayed in table 2 as model 3, and the results are significant. Second, the independent variable (PSM) should be significantly associated with the mediators (job impact and job contact). This is denoted as path *a* in figure 1. The results under model 1 show a significant coefficient value for the PSM variable on job impact, but the results in model 2 for job contact show an insignificant coefficient value for PSM. Third, the mediators should be significantly related to the dependent variable (path *b* in figure 1). The results under model 4 show a significant coefficient value for job impact but not for job contact.

Finally, when the mediator is controlled (i.e., paths *a* and *b*), the relationship between the independent and dependent variables should become less or no longer significant (path *c* in figure 1). A full mediation exists when a significant PSM variable becomes non-significant with the addition of the mediator. Fully mediated effects are rare in practice. A more common occurrence is partial mediation in which the PSM variable remains significant but its coefficient value drops with the addition of the mediator. The results in model 4 show a drop in the coefficient value of PSM from 0.39 to 0.29 but the

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TABLE 2 *Mediating effects of perceived job impact and perceived job contact on the PSM–job satisfaction relationship*

Variable	Relational job characteristics		Job satisfaction	
	Job impact (path a_1) Model 1	Job contact (path a_2) Model 2	(path *c*) Model 3	(path *b*) Model 4
Gender	0.32 (0.18)	0.36 (0.20)	0.38 (0.15)*	0.28 (0.14)*
Age	0.01 (0.01)	0.00 (0.01)	0.02 (0.01)*	0.01 (0.01)*
Education	0.05 (0.06)	−0.11 (0.07)	−0.04 (0.05)	−0.04 (0.05)
Supervisory position	0.26 (0.17)	0.40 (0.19)*	0.03 (0.14)	−0.06 (0.14)
Length of service	0.00 (0.01)	−0.01 (0.01)	−0.01 (0.01)	−0.01 (0.01)
PSM	0.31 (0.11)**	0.19 (0.13)	0.39 (0.10)***	0.27 (0.09)**
Perceived job impact				0.23 (0.07)**
Perceived job contact				0.09 (0.06)
R^2	0.09	0.05	0.13	0.24
F value	3.43**	1.87	5.38***	8.68***

Notes: Unstandardized regression coefficients are reported. Standard errors are in parentheses.
Level of significance: *p < 0.05; **p < 0.01; ***p < 0.001.

variable remains significant. A Sobel test using the critical values suggested by MacKinnon *et al.* (2004) indicated that the reduction in the effect of PSM on job satisfaction after the inclusion of job impact was statistically significant ($z = 2.26$, $p = 0.01$). The significant results suggest a partial mediating effect by job impact, but no mediating effect by job contact.

Moderation is determined in three steps. First, the effects of the independent variable (PSM) and moderator variables (job impact and job contact) on the dependent variable (job satisfaction) are examined (step 1). This is to establish whether the PSM, job impact, and job contact variables have statistically significant relationships with the job satisfaction variable. Second, the interaction between the PSM variable and each moderator variable is included to capture the combined effect of both. Step 2 examines the interaction between PSM and job impact, while step 3 examines the interaction between PSM and job contact. If the interaction is statistically significant, the difference in the PSM variable's effect between the first and second steps can be attributed to the moderator effect. The results in table 3 did not show any significant moderating effect by either job impact or job contact.

DISCUSSION

The primary purpose of this study is to examine whether a group of Australian local government workers' perceptions of two relational job characteristics have a mediating and/or moderating effect or negligible effect on the relationship between PSM and job satisfaction. The respondents' perceptions of job impact were found to partially mediate the association between PSM and job satisfaction. Earlier, Vinzant (1998) found higher job satisfaction among American protective service workers who thought that they had succeeded in helping people. She noted that some workers were dissatisfied with their job because they believed that they were not pursuing the common interest, and they blamed the 'system', their organization, and their organizational roles for preventing them from helping people.

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TABLE 3 *Moderating effects of perceived job impact and perceived job contact on the PSM–job satisfaction relationship*

Variable	Step 1	Step 2	Step 3
Gender	0.28 (0.14)*	0.28 (0.14)*	0.30 (0.14)*
Age	0.01 (0.01)*	0.01 (0.01)*	0.01 (0.01)*
Education	−0.04 (0.05)	−0.04 (0.05)	−0.04 (0.05)
Supervisory position	−0.06 (0.14)	−0.05 (0.13)	−0.04 (0.14)
Length of service	−0.01 (0.01)	−0.01 (0.01)	−0.01 (0.01)
PSM	0.19 (0.06)**	0.18 (0.06)**	0.19 (0.06)**
Perceived job impact	0.26 (0.08)**	0.26 (0.08)**	0.24 (0.09)**
Perceived job contact	0.12 (0.08)	0.11 (0.08)	0.13 (0.08)
PSM × perceived job impact		−0.02 (0.06)	0.04 (0.08)
PSM × perceived job contact			−0.08 (0.09)
R^2	0.24	0.24	0.24
F	8.62***	7.65***	6.98***

Notes: Unstandardized regression coefficients are reported. Standard errors are in parentheses.
Level of significance: *p < 0.05; **p < 0.01; ***p < 0.001.

This study not only supports prior findings that workers with high PSM levels tend to be more satisfied with their job than those with low PSM levels (Taylor 2007; Pandey and Stazyk 2008), but also suggests that government employees who have strong norms about performing public service were more satisfied with their job, partly based on the extent to which they perceived that their jobs provided avenues for worthwhile accomplishment through the frequency, magnitude, and scope of job impact. High PSM workers are generally more concerned about doing work that benefits others (Grant and Wrzesniewski 2010), and show a stronger tendency to self-monitor the progress of their work (Weber *et al.* 2004) than workers with low PSM levels. These workers with high PSM levels are likely to be more satisfied with their job when they are convinced that their jobs allow them to have a positive impact on citizens (Knoke and Wright-Isak 1982; Kristof 1996).

This finding suggests the importance of empowering employees in the delivery of public service so that they can personally witness the positive impact of their efforts on the well-being of others. There are likely to be some government jobs that are high in job impact and others that are low in job impact. Helping employees see that they are meaningfully contributing to organizational goals can reduce their frustration and strengthen their commitment (Romzek and Hendricks 1982). If employees with high PSM levels are more concerned about making a positive difference than those with low PSM levels, then job impact cues and adequate support from their organization could communicate to them about their positive contributions. While it is clearly important that managers should communicate to their subordinates about 'the centrality of their role in the organization and the real benefits that their contribution makes to society' (Moynihan and Pandey 2007, p. 48), this study also suggests the value of facilitating employees to see for themselves that their actions matter in other people's lives.

Does this finding also suggest that low PSM workers will show low satisfaction levels because of perceptions of low job impact? Based on the finding that high PSM employees generally report high job satisfaction partly because of perceptions of high job impact, it would be tempting to deduce the opposite effect for low PSM workers. After all, high task

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significance or job impact had been reported to have a positive impact on job outcome (Hackman and Oldham 1976). On the other hand, if employees with low PSM levels are less concerned about the consequences of their actions on others (Korsgaard *et al.* 1997) and are less likely to self-monitor their work (Weber *et al.* 2004), then they are likely to be less responsive to the presence of low job impact. It is possible for their job satisfaction levels to be less influenced by the presence of low job impact than their counterparts with high PSM levels. Since job satisfaction reflects the extent to which employees' job needs are fulfilled in their organization, the satisfaction level of an employee with a low PSM level can be determined more by the factors that they value rather than by a chance to do good. Rather than assume that low PSM workers would respond in a manner that is the direct contrast to that of high PSM workers, an experimental study can be designed to compare the responses of these two groups of workers to different levels of job impact.

In contrast to job impact, the frequency and breadth of job contact with service recipients did not show a mediating or moderating effect on the PSM–job satisfaction relationship. In fact, job contact with citizens was not found to be significantly associated with PSM, which conflicts with earlier findings (Grant 2007, 2008b). While the relational job design literature views the interactions with service recipients to be a pleasant experience (Grant 2007; Humphrey *et al.* 2007), Lipsky's (1980) study on street-level bureaucrats and the emotional labour literature suggest that interactions with service recipients involve challenges and tensions. This study found that the intensity and breadth of job contact with service recipients did not significantly influence PSM, and the association between PSM and job satisfaction. One possible interpretation of this finding is that there could be other factors associated with job contact that matter more to the respondents than just frequency and breadth of job contact, such as the contact experience. Citizen interactions that are characterized by respect and appreciation, as opposed to those that involve disproportionate or ambiguous expectations and unfriendly behaviours or verbal aggression can produce different outcomes. The employees' occupation and organizational role could also matter. Further research should include these factors when examining the effects of contact with citizens.

It should be noted that this study's cross-sectional design does not permit causal relationships between study variables and the interaction effects to be determined with confidence. Although the directions of the causal relationships are informed by the literature, it is possible for reverse causality to occur. Drawing from self-efficacy research, Wright and Grant (2010) argued that when employees perform their task effectively, they are likely to experience greater confidence in their capabilities to perform their civic duties, which in turn strengthens their public service motives. Further, Weber *et al.* (2004) pointed out that March and Olsen's (1989) logic of appropriateness framework proposes not only that identity factors interact with organizational cues to determine an individual's definition of the organizational situation but also that organizational cues could elicit the identity factors that then interact with the organizational cues. Experimental data would yield better internal validity (McClelland and Judd 1993).

Further PSM studies on the job satisfaction of government officials should incorporate alternative designs and include salary as a control variable. Although the two post-hoc statistical techniques for detecting CMB indicate that it does not appear to be a pervasive problem in this study, they are subject to limitations (Podsakoff *et al.* 2003, 2012). It would be more prudent to minimize CMB in the first place by collecting data from more than one source. Finally, like PSM, salary is closely linked to public sector job satisfaction (Herzberg

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1966), and is reported to be important for many government employees (Rainey 1997). Future studies on PSM's effects on job satisfaction should include it as a control variable.

Although this study found that high PSM leads to high job satisfaction, it does not ignore the possibility of unfavourable outcomes. In their study of Swiss local government workers, Giauque *et al.* (2011) noted that two dimensions of PSM – compassion and self-sacrifice – induced resignation. They concluded that some public service orientations of employees may contribute to expectations that are incompatible with the working conditions and operations of public sector organizations. They also suggested that employees who are frustrated by their inability to do good in their work environment but continue to work in the organization could experience negative effects on their psychological well-being.

Negative effects such as reduced job satisfaction and burnout are particularly prevalent in jobs with high impact that have the potential to harm some service recipients (Grant and Campbell 2007). For example, physicians were found to experience distress after causing medical errors that harm patients (Hammer 1985). Harming their service recipients can cause employees with high PSM levels to experience guilt. Their attempts to manage the guilt, coupled with the anxiety about causing further harm to the service recipients, can contribute to emotional exhaustion (Molinsky and Margolis 2005). To protect themselves from being overwhelmed by these feelings, employees can psychologically distance and detach themselves from the interpersonal relationships with the service recipients, leading to a sense of depersonalization (Folger and Skarlicki 1998; Molinsky and Margolis 2005).

It is apparent that PSM has the potential to produce positive and negative effects, and that the job outcomes of PSM can be influenced in different ways by various situational factors. This research on a group of Australian local government workers reinforces the importance of incorporating different organizational factors when examining the effects of PSM on job outcomes in order to harness the positive effects and minimize any negative effects of PSM in the public sector workplace.

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APPENDIX 1: SURVEY SCALES AND ITEMS

Public service motivation	
Compassion	(COM1) I feel sympathetic to the plight of the underprivileged.
	(COM2) I get very upset when I see other people being treated unfairly.
Commitment to public values	(CPI2) It is important for me to contribute to the common good.
	(CPI1) Meaningful public service is very important to me.
Self-sacrifice	(SS1) I am prepared to make sacrifices for the good of society.
	(SS2) People should give back to society more than what they get from it.
Attraction to public service	(CPV1) It is fundamental that the interests of future generations are taken into account when developing public policies.
Job satisfaction	
	(JS1) I am fairly well satisfied with my present job.
	(JS2) Most days I am enthusiastic about my work.
	(JS3) I like my job better than the average worker does.
	(JS4) I feel enjoyment in my work.
Perceived job impact	
Magnitude	(JIM1) My job provides opportunities to improve the welfare of others.
Frequency	(JIF1) My job allows me to have a positive impact on others almost every day.
	(JIF2) My job frequently improves the lives of others.
Scope	(JIS1) A lot of people can be positively affected by how well my job gets done.
	(JIS2) My job provides opportunities to have a positive impact on a large number of other people.
Perceived job contact	
Frequency	(JCF1) My job allows frequent communication with the people who benefit from my work.
	(JCF2) My job often gives me the opportunity to meet the people who benefit from my work.
Breadth	(JCB1) My job provides me with contact with different groups of people who benefit from my work.
	(JCB2) My job allows me to interact with a variety of people who benefit from my work.

Control variables
Gender (1 = male; 2 = female)
Age (in years)
Educational qualifications (1 = high school; 2 = vocational education; 3 = other certificate or diploma, excluding a university degree; 4 = university undergraduate or postgraduate degree)
Supervisory position (1 = non-supervisory; 2 = supervisory)
Length of service (in years).

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APPENDIX 2: FACTOR LOADINGS OF THE SURVEY ITEMS

Scales	Items	Factor 1	Factor 2	Factor 3	Factor 4
Job impact	JIF2	**0.91**	0.60	0.21	0.37
	JIM1	**0.86**	0.61	0.18	0.37
	JIF1	**0.86**	0.63	0.21	0.38
	JIS2	**0.83**	0.53	0.24	0.35
	JIS1	**0.76**	0.55	0.28	0.37
Job contact	JCB2	0.63	**0.96**	0.12	0.33
	JCF2	0.63	**0.92**	0.11	0.33
	JCB1	0.65	**0.91**	0.10	0.29
	JCF1	0.62	**0.88**	0.10	0.36
PSM	CPI2	0.19	0.11	**0.88**	0.21
	CPI1	0.26	0.19	**0.73**	0.29
	SS1	0.13	0.05	**0.56**	0.17
	COM1	0.08	0.03	**0.51**	0.11
	COM2	0.20	0.04	**0.48**	0.21
	SS2	−0.02	−0.04	**0.45**	0.10
	CPV1	0.23	0.09	**0.43**	0.14
Job satisfaction	JS1	0.36	0.30	0.32	**0.80**
	JS4	0.36	0.31	0.23	**0.76**
	JS2	0.24	0.19	0.13	**0.76**
	JS3	0.36	0.29	0.26	**0.68**

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